Organizational Structures

Lecture 5

ASTANA IT UNIVERSITY

Introduction

More than forty years ago, Wallace identified four major factors that caused the onset of the organizational revolution:

- The technology revolution (complexity and variety of products, new materials and processes, and the effects of massive research)
- Competition and the profit squeeze (saturated markets, inflation of wage and material costs, and production efficiency)
- The high cost of marketing
- The unpredictability of consumer demands (due to high income, wide range of choices available, and shifting tastes)

a B 8 a B T B 8 B B x B w B B 4 B 4 3), pp. 2 3 3 O 2 B . m B 3 B 1 B w T B) B d l i

ASTANA IT UNIVERSITY

Introduction

According to Grinnell and Apple, there are five general indications that the traditional structure may not be adequate for managing projects:

- Management is satisfied with its technical skills, but projects are not meeting time, cost, and other project requirements.
- There is a high commitment to getting project work done, but great fluctuations in how well performance specifications are met.
- Highly talented specialists involved in the project feel exploited and misused.
- Particular technical groups or individuals constantly blame each other for failure to meet specifications or delivery dates.
- Projects are on time and to specifications, but groups and individuals a r e n ' satisfied with the achievement.

7 a B 6 a B , B B . a B 4 a B u *Machine Design*, January 1975, pp. 84 87. B u B v B 8 B 3 T B

ASTANA IT UNIVERSITY

Introduction

According to Sadler:

Since the relative influence of these (sociotechnical) factors change from situation to situation, there can be no such thing as an ideal structure making for effectiveness in organizations of all kinds, or even appropriate to a single type of organization at different stages in its development.

4 B 7 T B x B B 3 B 7 33. T B 1 B (B 6 T B a B d d T B 2 a B i T B d l j d T E

Introduction

Sadler has prepared a six-question checklist that explores a company's task, social climate, and relationship to the environment.

- To what extent does the task of organization call for close control if it is to be performed efficiently?

- What are the needs and attitudes of the people performing the tasks? What are the likely effects of control mechanisms on their motivation and performance?

- What are the natural social groupings with which people identify themselves? To what extent are satisfying social relationships important in relation to motivation and performance?

4 B 7 T B x B B 3 B 7 33. T B 1 B (B 6 T B a B d d T B 2 a B i T B d l j d T E

Introduction

- What aspect of the organization's activities needs to be closely integrated if the overall task is to be achieved?

- What organizational measures can be developed that will provide an appropriate measure of control and integration of work activities, while at the same time meeting the needs of people and providing adequate motivation?

- What environmental changes are likely to affect the future trend of company operations? What organizational measures can be taken to insure that the enterprise responds to these effectively?

4 B7 TB x B B3 B7 33. T B1 B(B6 TB aBddTB2 aBiTBdljdTB

Organizational Work Flow

In the discussion of organizational structures, the following definitions will be used:

- Authority is the power granted to individuals (possibly by their position) so that they can make final decisions.

- Responsibility is the obligation incurred by individuals in their roles in the formal organization to effectively perform assignments.

- Accountability is being answerable for the satisfactory completion of a specific assignment. (Accountability = authority + responsibility.)

ASTANA IT UNIVERSITY

Organizational Work Flow

Trust is the key to success here. The normal progression in the growth of trust is as follows:

- Even though a problem exists, both the project and functional managers deny that any problem exists.
- When the problem finally surfaces, each manager blames the other.
- As trust develops, both managers readily admit responsibility for the problems.
- The project and functional managers meet face-to-face to work out the problem.
- The project and functional managers begin to formally and informally anticipate problems.

ASTANA IT UNIVERSITY

Traditional (Classical) Organization

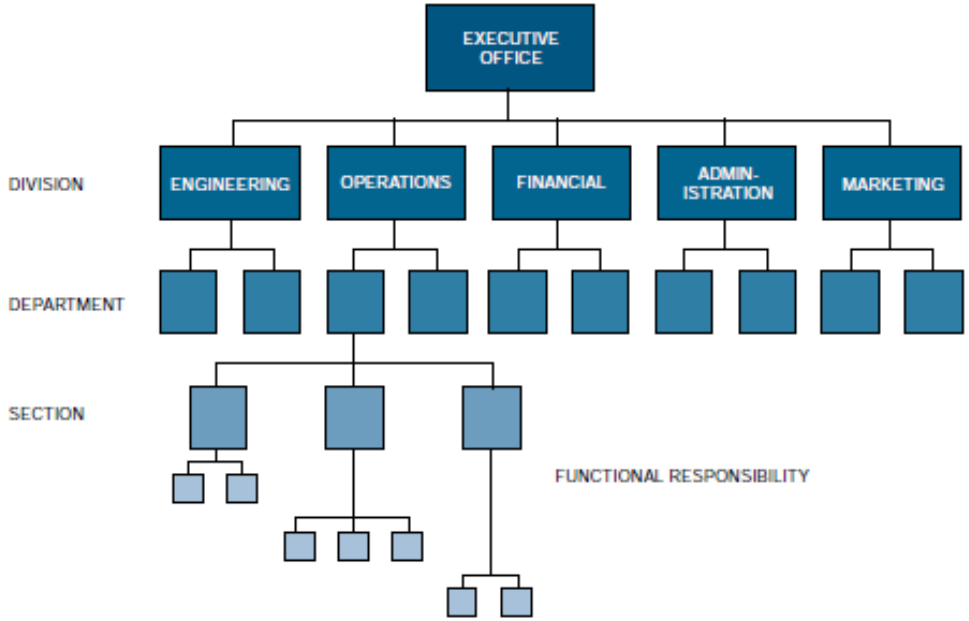


Fig. 1. The traditional management structure

ASTANA IT UNIVERSITY

Traditional (Classical) Organization

Advantages of the Traditional (Classical) Organization

- Easier budgeting and cost control are possible.
- Better technical control is possible.
- Specialists can be grouped to share knowledge and responsibility.
- Personnel can be used on many different projects.
- All projects will benefit from the most advanced technology (better utilization of scarce personnel).

ASTANA IT UNIVERSITY

Traditional (Classical) Organization

Advantages of the Traditional (Classical) Organization (continued)

- A broad manpower base to work with.
- Continuity in the functional disciplines; policies, procedures, and lines of responsibility are easily defined and understandable.
- Admits mass production activities within established specifications.
- Good control over personnel, since each employee has one and only one person to report to.
- Communication channels are vertical and well established.
- Quick reaction capability exists, but may be dependent upon the priorities of the functional managers.

ASTANA IT UNIVERSITY

Traditional (Classical) Organization

Disadvantages of the Traditional (Classical) Organization

- No one individual is directly responsible for the total project (i.e., no formal authority; committee solutions).

- Does not provide the project-oriented emphasis necessary to accomplish the project tasks.

- Coordination becomes complex, and additional lead time is required for approval of decisions.

- Decisions normally favor the strongest functional groups.

ASTANA IT UNIVERSITY

Traditional (Classical) Organization

Disadvantages of the Traditional (Classical) Organization (continued)

- No customer focal point.
- Response to customer needs is slow.
- Difficulty in pinpointing responsibility; this is the result of little or no direct project reporting, very little project-oriented planning, and no project authority.
- Motivation and innovation are decreased.
- Ideas tend to be functionally oriented with little regard for ongoing projects.

ASTANA IT UNIVERSITY

Traditional (Classical) Organization

Projects have a tendency to fall behind schedule in the classical organizational structure.

Incredibly large lead times are required. Functional managers attend to those tasks that provide better benefits to themselves and their subordinates first.

With the growth of project management in the late 1960s, executives began to realize that many of the problems were the result of weaknesses in the traditional structure.

ASTANA IT UNIVERSITY

Developing Work Integration Positions

As companies grew in size, more emphasis was placed on multiple ongoing programs with high-technology requirements.

Organizational pitfalls soon appeared, especially in the integration of the flow of work.

As management discovered that the critical point in any program is the interface between functional units, the new theories of "integration management" developed.

Because of the interfacing problems, management began searching for innovative methods to coordinate the flow of work between functional units without modification to the existing organizational structure.

ASTANA IT UNIVERSITY

Developing Work Integration Positions

This coordination was achieved through several integrating mechanisms:

- Rules and procedures
- Planning processes
- Hierarchical referral
- Direct contact

) B6aB, TB 1 B3 Bx a B6 40. B B B Bv B. TBz Bdlj

Developing Work Integration Positions

When the need for project managers was acknowledged, the next logical question was where in the organization to place them.

Executives preferred to keep project managers low in the organization.

After all, if they reported to someone high up, they would have to be paid more and would pose a continuous threat to management.

The first attempt to resolve this problem was to develop project leaders or coordinators within each functional department.

Section-level personnel were temporarily assigned as project leaders and would return to their former positions at project termination.

This is why the term "project leader" is used rather than "project manager," which implies a permanent relationship.

ASTANA IT UNIVERSITY

Developing Work Integration Positions

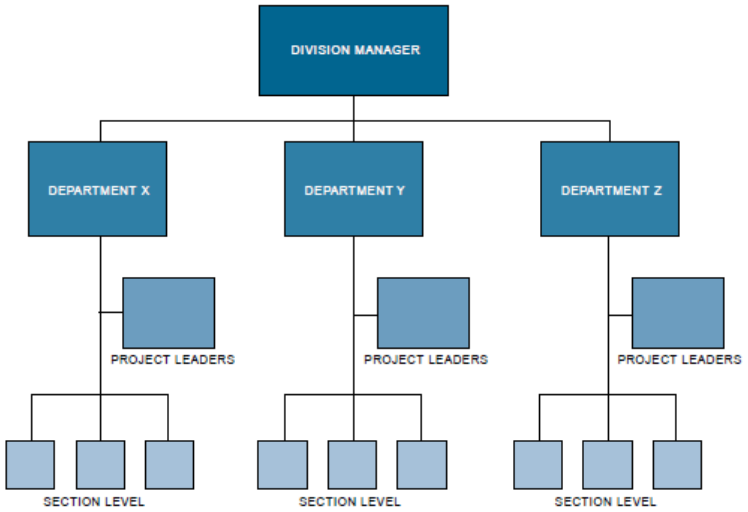


Fig. 2. Departmental project management

ASTANA IT UNIVERSITY

Developing Work Integration Positions

The next step in the evolution of project management was the task force concept.

The rationale behind the task force concept was that integration could be achieved if each functional unit placed a representative on the task force.

The group could then jointly solve problems as they occurred, provided that budget limitations were still adhered to.

Theoretically, decisions could now be made at the lowest possible levels, thus expediting information and reducing, or even eliminating, delay time.

ASTANA IT UNIVERSITY

Developing Work Integration Positions

Development of the task force concept was a giant step toward conflict resolution: Work was being accomplished on time, schedules were being maintained, and costs were usually within budget.

However, without formal project authority relationships, task force members remained loyal to their functional organizations, and when conflicts came about between the project and functional organization, the project always suffered.

Although the task force concept was a step in the right direction, the disadvantages strongly outweighed the advantages.

ASTANA IT UNIVERSITY

Developing Work Integration Positions

The next step in the evolution of work integration was the establishment of liaison departments, particularly in engineering divisions that perform multiple projects involving a high level of technology (see Figure 3). The purpose of the liaison department was to handle transactions between functional units within the (engineering) division.

The liaison personnel received their authority through the division head. The liaison department did not actually worked toward the same requirements and goals. Liaison departments are still in existence in many large companies and typically handle engineering changes and design problems.

ASTANA IT UNIVERSITY

Developing Work Integration Positions

Unfortunately, the liaison department is simply a scale up of the project coordinator within the department. The authority given to the liaison department extends only to the outer boundaries of the division.

If a conflict arose between the manufacturing and engineering divisions, for example, it would still be referred to upper management for resolution.

Today, liaison departments are synonymous with project engineering and systems engineering departments, and the individuals in these departments have the authority to span the entire organization.

ASTANA IT UNIVERSITY

Developing Work Integration Positions

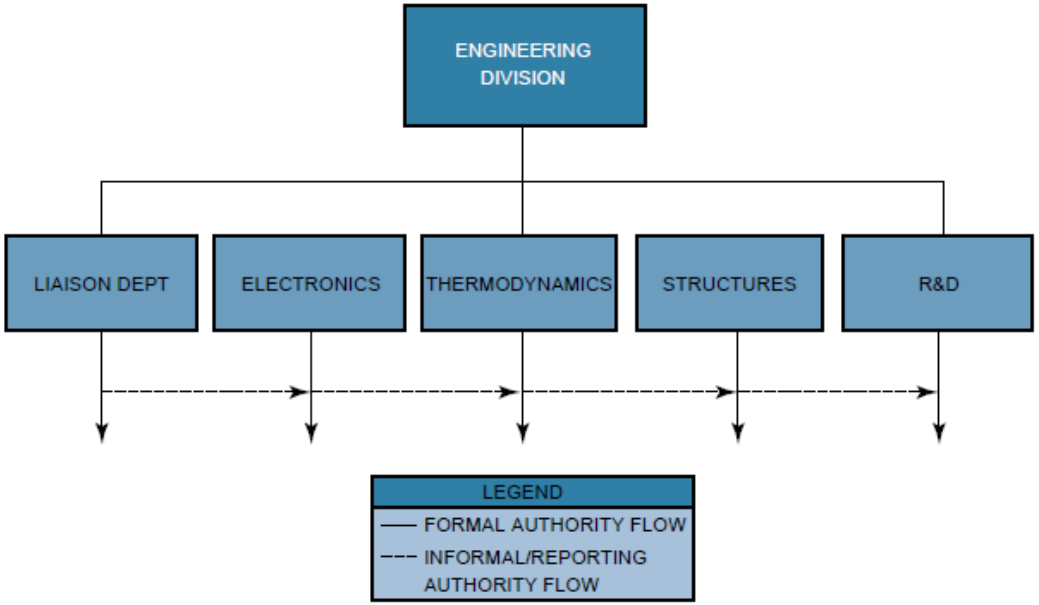


Fig. 3. Engineering division with liaison department (The Expeditor)

ASTANA IT UNIVERSITY

Developing Work Integration Positions

It soon became obvious that control of a project must be given to personnel whose first loyalty is directed toward the completion of the project.

Thus the project management position must not be controlled by the functional managers. Figure 4 shows a typical line –staff organization.

ASTANA IT UNIVERSITY

Developing Work Integration Positions

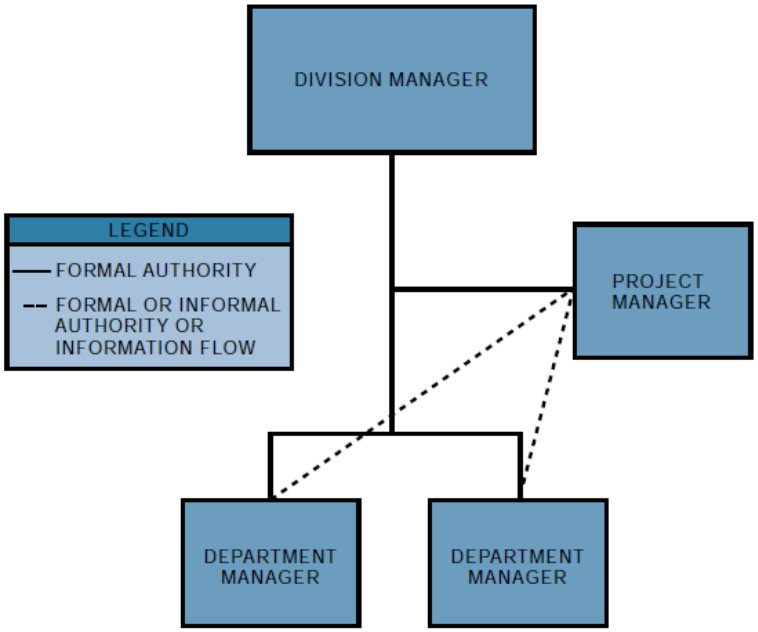


Fig. 4. Line –staff organization (Project Coordinator)

ASTANA IT UNIVERSITY

Pure Product (Projectized) Organization

The pure product organization, as shown in Figure 5, develops as a division within a division.

As long as there exists a continuous flow of projects, work is stable, and conflicts are at a minimum.

The major advantage of this organizational flow is that one individual, the program manager, maintains complete line authority over the entire project.

Not only does he assign work, but he also conducts merit reviews.

ASTANA IT UNIVERSITY

Pure Product (Projectized) Organization

The major disadvantage with the pure project form is the cost of maintaining the organization.

There is no chance for sharing an individual with another project in order to reduce costs.

Personnel are usually attached to these projects long after they are needed because once an employee is given up, the project manager might not be able to get him back.

ASTANA IT UNIVERSITY

Pure Product (Projectized) Organization

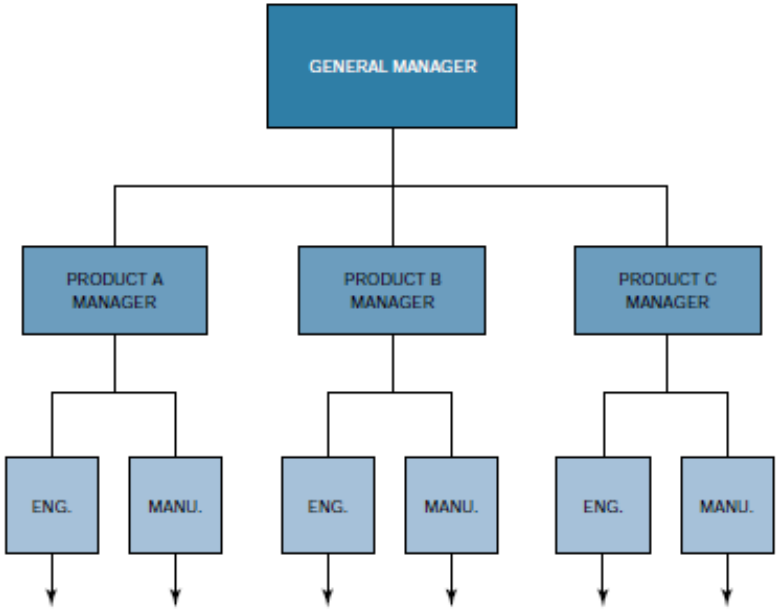


Fig. 5. Pure product or projectized structure

ASTANA IT UNIVERSITY

Pure Product (Projectized) Organization

- Provides complete line authority over the project (i.e., strong control through a single project authority).
- Participants work directly for the project manager. Unprofitable product lines are easily identified and can be eliminated.
- Strong communications channels.
- Staffs can maintain expertise on a given project without sharing key personnel.
- Very rapid reaction time is provided.
- Personnel demonstrate loyalty to the project; better morale with product identification.
- A focal point develops for out-of-company customer relations.
- Flexibility in determining time (schedule), cost, and performance trade-offs.
- Interface management becomes easier as unit size is decreased.
- Upper-level management maintains more free time for executive decision-making.

Table 1. Advantages of the Product Organizational Form

ASTANA IT UNIVERSITY

Pure Product (Projectized) Organization

- Cost of maintaining this form in a multiproduct company would be prohibitive due to duplication of effort, facilities, and personnel; inefficient usage.
- A tendency to retain personnel on a project long after they are needed. Upper-level management must balance workloads as projects start up and are phased out.
- Technology suffers because, without strong functional groups, outlook of the future to improve company's capabilities for new programs would be hampered (i.e., no perpetuation of technology).
- Control of functional (i.e., organizational) specialists requires top-level coordination.
- Lack of opportunities for technical interchange between projects.
- Lack of career continuity and opportunities for project personnel.

Table 2. Disadvantages of the Product Organizational Form

ASTANA IT UNIVERSITY

Matrix Organizational Form

The matrix organizational form is an attempt to combine the advantages of the pure functional structure and the product organizational structure. This form is ideally suited for companies, such as construction, that are "project driven."

Since each project represents a potential profit center, the power and authority used by the project manager come directly from the general manager. The project manager has total responsibility and accountability for project success. The functional departments, on the other hand, have functional responsibility to maintain technical excellence on the project.

ASTANA IT UNIVERSITY

Matrix Organizational Form

The matrix structure therefore becomes a compromise in an attempt to obtain the best of two worlds. In pure product management, technology suffered because there was a single group for planning and integration.

In the pure functional organization, time and schedule were sacrificed.

Matrix project management is an attempt to obtain maximum technology and performance in a cost-effective manner and within time and schedule constraints.

ASTANA IT UNIVERSITY

Matrix Organizational Form

Obviously, the matrix structure is the most complex of all organizational forms. Grinnell and Apple define four situations where it is most practical to consider a matrix:

- When complex, short-run products are the organization's primary output.
- When a complicated design calls for both innovation and timely completion.
- When several kinds of sophisticated skills are needed in designing, building, and testing the products—skills then need constant updating and development.
- When a rapidly changing marketplace calls for significant changes in products, perhaps between the time they are conceived and delivered.

ASTANA IT UNIVERSITY

Matrix Organizational Form

Matrix implementation requires:

- Training in matrix operations
- Training in how to maintain open communications
- Training in problem solving
- Compatible reward systems
- Role definitions

ASTANA IT UNIVERSITY

Modification of Matrix Structures

The matrix can take many forms, but there are basically three common varieties. Each type represents a different degree of authority attributed to the program manager and indirectly identifies the relative size of the company.

As an example, in the matrix of previous Figure 5, all program managers report directly to the general manager. This type of arrangement works best for small companies that have few projects and assumes that the general manager has sufficient time to coordinate activities between his project managers. In this type of arrangement, all conflicts between projects are referred to the general manager for resolution.

ASTANA IT UNIVERSITY

Modification of Matrix Structures

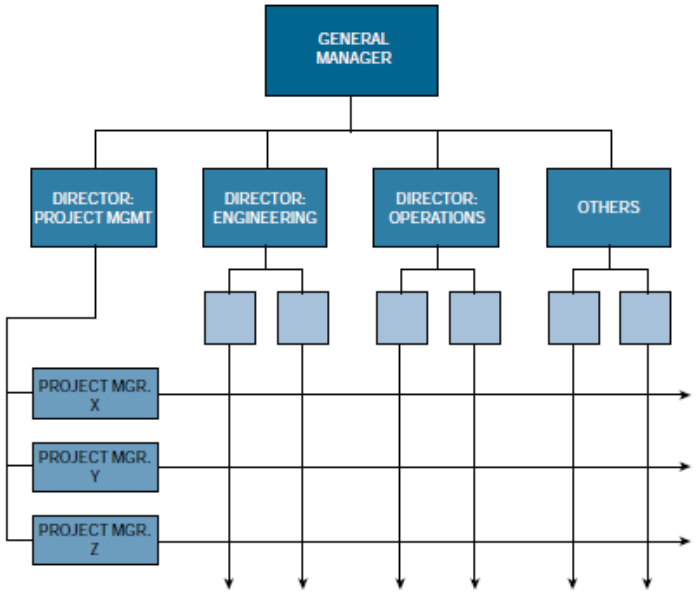


Fig. 6. Development of a director of project management

ASTANA IT UNIVERSITY

Modification of Matrix Structures

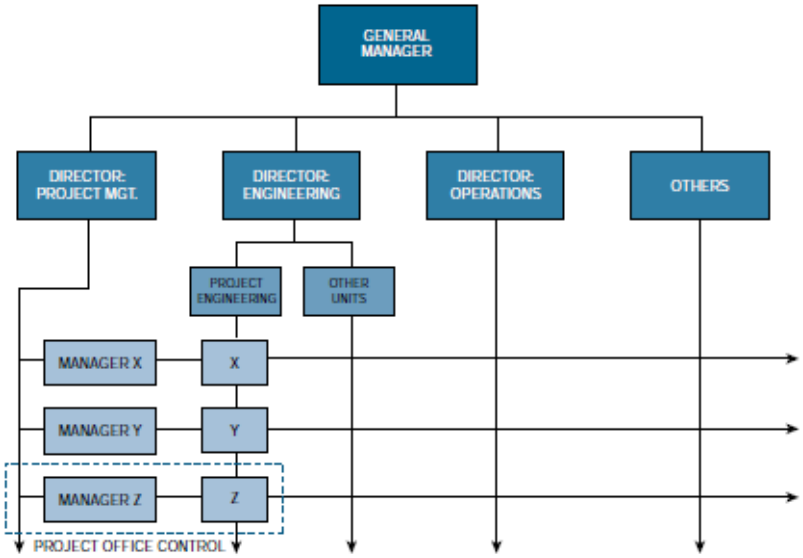


Fig. 7. Placing project engineering in the project office

ASTANA IT UNIVERSITY

Modification of Matrix Structures

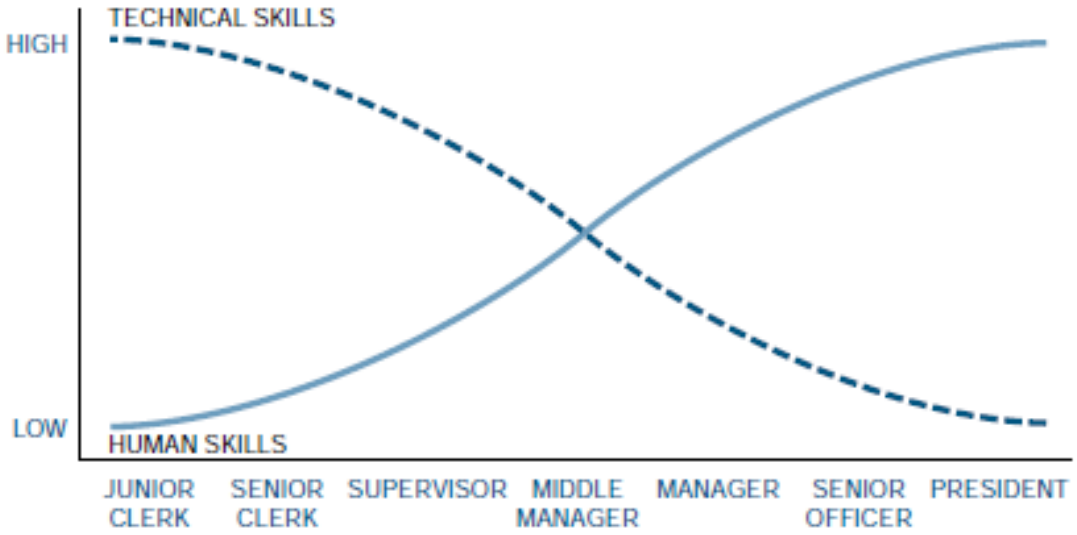


Fig. 8. Philosophy of management

ASTANA IT UNIVERSITY

Modification of Matrix Structures

Project Management	*Project Engineering*
• Total project planning • Cost control • Schedule control • System specifications • Logistics support	• Total project planning • Cost control • Schedule control • System specifications • Logistics support
• Contract control • Report preparation and distribution • Procurement • Identification of reliability and maintainability requirements • Staffing • Priority scheduling • Management information systems	• Configuration control • Fabrication, testing, and production technical leadership support

Table 3. Project Management Compared to Project Engineering

ASTANA IT UNIVERSITY

Modification of Matrix Structures

The main reason for separating project management from project engineering is so that the project engineer will remain "solid" with the director of engineering in order to have the full authority to give technical direction to engineering.

ASTANA IT UNIVERSITY

The Strong, Weak, or Balanced Matrix

Matrix structures can be strong, weak, or balanced.

The strength of the matrix is based upon who has more influence over the daily performance of the workers: project manager or line managers.

If the project manager has more influence over the worker, then the matrix structure functions as a strong matrix as seen through the eyes of the project manager.

If the line manager has more influence than does the project manager, then the organization functions as a weak matrix as seen by the project manager.

ASTANA IT UNIVERSITY

Reference

Project Management: A Systems Approach to Planning, Scheduling, and Controlling, 12th ed. Harold Kerzner, ISBN-10: 9781119165354, ISBN-13: 978-1119165354, 2017

S. K. Grinnell and H. P. Apple, "One," Machine Design, vol. 37, January

John M. Stewart, "Making Proje
with permission from Business Horizons, Fall 1965 (p. 54). Copyright © 1964 by the Board of Trustees at Indiana University.

ASTANA IT UNIVERSITY